nd2_maxip: User Guide

Turn Nikon .nd2 files into max-projection TIFF images, one per multipoint position. No Python experience needed.

What this program does

Nikon AX confocal files (.nd2) can hold many stage positions (multipoints), each with a z-stack. This program:

- Splits the file into its individual positions.
- Collapses each z-stack into one flat image using a **maximum intensity projection** (the brightest value at each pixel across all z-slices).
- Saves each position as its own .tif file.

It can save two kinds of TIFF:

Type	Best for	Details
Standard (default)	Analysis in Fiji / ImageJ	Keeps the original bit depth and every channel as a separate layer. Pixel size is stored in the file.
Simple (`--simple`)	PowerPoint, Windows Photos, email	A normal color picture (8-bit RGB) with all channels merged, using each channel's color. Not suitable for measuring intensities.

One-time setup

Step 1. Install Python. Download it from python.org/downloads and run the installer. **On the first screen, tick the box "Add python.exe to PATH"** before clicking Install. This box is easy to miss and it prevents most setup problems.

Step 2. Open a command window. On Windows, press the Windows key, type `cmd`, and press Enter. (On a Mac, open the Terminal app.) This is where you will type the commands below.

Step 3. Install the required libraries. Libraries are add-on packages the program relies on. Copy this line into the command window and press Enter:

```
pip install nd2 tifffile numpy dask
```

Library	What it is used for
nd2	Reads Nikon .nd2 files
tifffile	Writes the .tif files
numpy	Does the image math (the projection)
dask	Lets large files load piece by piece instead of all at once

You only need to do this once per computer. It may take a minute and print a lot of text; that is normal.

Step 4. Save the program. Put `nd2_maxip.py` in any folder you like, for example `C:\nd2tools`.

Running the program

1. Go to the folder with the program. In the command window, type `cd` followed by a space, then drag the folder from File Explorer into the window (this fills in the path for you) and press Enter:

```
cd C:\nd2tools
```

2. Run it on a file. Use quotes around any path that contains spaces:

```
python nd2_maxip.py "C:\data\experiment1.nd2"
```

If `python` is not recognized, try `py` instead, or see Troubleshooting below.

The TIFFs are saved in the same folder as the .nd2 file, one per position. Each name has the file name, the position number, and the position name that was set in NIS-Elements, for example `experiment1_P001_WellA1.tif` and `experiment1_P002_Control.tif`. Characters Windows does not allow in file names (such as / : ?) are replaced with underscores, and spaces become underscores. If a position has no name, only the number is used.

Common examples

I want to...	Type this
Process one file	`python nd2_maxip.py "C:\data\a.nd2"`
Process every .nd2 in a folder	`python nd2_maxip.py "C:\data\run1"`
Save results to a different folder	`python nd2_maxip.py "C:\data\a.nd2" -o "C:\data\output"`
Make PowerPoint-friendly color images	`python nd2_maxip.py "C:\data\a.nd2" --simple`
Replace files that already exist	`python nd2_maxip.py "C:\data\a.nd2" --overwrite`

All options

Option	What it does
`-o FOLDER`	Save the TIFFs in FOLDER instead of next to the .nd2 file. The folder is created if needed.
`--overwrite`	Replace TIFFs that already exist. Without this, existing files are skipped.
`--simple`	Save 8-bit color RGB TIFFs (file names end in `_rgb.tif`). If the file has a time series, you get one TIFF per time point (`_T001`, `_T002`, ...).
`--colors blue,red`	With --simple: choose the color of each channel, in channel order. Names: red, green, blue, cyan, magenta, yellow, orange, gray. Hex codes like `#0000ff` also work. Normally not needed, because colors are read from the file.
`--low 0.5` `--high 99.8`	With --simple: controls brightness/contrast. The image is stretched between these two percentiles. Raise --low or lower --high for more contrast; do the opposite if bright areas look washed out.

When --simple runs, it prints the channel colors it found, for example `channel colors from file (RGB): [(0, 0, 255), (255, 0, 0)]`. That is blue then red. Check that this matches what you expect.

Good to know

- **Keep the standard TIFFs for measurements.** The --simple images are adjusted for display, so their brightness values are no longer the original data.
- The original .nd2 file is never changed.
- Opening a standard TIFF in Fiji: use File > Open. Channels appear as a slider at the bottom of the image.

Troubleshooting

Problem	What to try
`'python' is not recognized`	Try `py` in place of `python`. If that fails, reinstall Python and tick "Add python.exe to PATH".
`No module named 'nd2'` (or tifffile, numpy, dask)	The libraries are not installed. Run the `pip install` line from Step 3 again. If `pip` is not recognized, use `python -m pip install nd2 tifffile numpy dask`.
`No .nd2 files found.`	The path is wrong or missing quotes. Check the spelling and put quotes around the path.
Skipped: "exists; use --overwrite"	Output files with that name already exist. Add `--overwrite` or choose a new folder with `-o`.
Warning: "could not read channel colors"	The file's colors could not be read, so default colors were used. Set them yourself with `--colors`, for example `--colors blue,red`.

Problem	What to try
Channel colors look wrong	Use `--colors` to set them manually.
Out-of-memory error or very slow	Close other programs and process one file at a time.